

MOSAIC: A Dual-Layer Fault-Tolerant Molecular Storage Codec with Empirical Coverage Threshold Characterisation for Nanopore Sequencing

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Abstract—Ultra-high-density storage media offer unprecedented theoretical capacities, yet their commercial viability is bottlenecked by the stochastic, error-prone nature of their read channels. In this paper, we introduce **MOSAIC** (Molecular Object Storage Architecture via Independent Coding), a dual-state, fault-tolerant storage architecture designed to decouple data integrity from underlying physical media degradation. Utilizing synthetic DNA and nanopore sequencing as a representative high-error physical layer, we demonstrate how data whitening and block-level erasure coding maximize survivability against severe insertion, deletion, and substitution (IDS) errors. While classical algebraic error correction excels at localized substitutions and erasures, these legacy architectures experience unrecoverable desynchronization under the severe asynchronous frame-shifting (indels) native to molecular channels. We evaluate MOSAIC in two operational states: *Half* MOSAIC, a pragmatic baseline implementing inner-sequence GF(2⁸) Reed-Solomon correction for density-sensitive multimedia payloads, and *Full* MOSAIC, a hardened architecture that integrates a global Reed-Solomon erasure matrix for deterministic, block-level integrity.

Rather than relying on the graceful degradation of consensus-only methods—which permits silent data corruption—*Full* MOSAIC enforces a strict, block-level failure budget ($T = 32$ erasures/block), ensuring SHA-256-verified recovery. Through an empirical ablation study across legacy (R9.4.1) and next-generation (R10.4) physical channels on payloads up to 1 MB, we demonstrate the unrecoverable decoding failure modes of single-layer consensus. *Full* MOSAIC achieves exact recovery convergence at $4\times$ – $5\times$ coverage on R10.4 and $10\times$ – $12\times$ on R9.4 channels. This reduces the minimum required channel oversampling by over 80% relative to consensus-only approaches, establishing a reliable SLA for cost-efficient large-scale archival systems.

Index Terms—Fault-tolerant storage, Reed-Solomon coding, data whitening, channel coding, nanopore sequencing, DNA data storage, erasure codes, systems architecture.

1 INTRODUCTION

SYNTHETIC DNA has emerged as a highly viable medium for cold-tier large-scale archiving, offering a theoretical volumetric density exceeding 10^{18} bytes per gram and near-permanent stability under controlled conditions [3], [4].

However, commercial deployment is fundamentally constrained by the high-error nanopore profiles of read channels, which behave as stochastic systems dominated by severe insertions, deletions, and substitutions (IDS). To mitigate this channel noise, current state-of-the-art architectures rely almost exclusively on algorithmic read consensus, typically requiring physical read oversampling exceeding $20\times$ to probabilistically recover the master sequence [5].

This high-oversampling paradigm introduces a severe economic bottleneck: read depth scales linearly with retrieval cost. More critically, consensus-only architectures exhibit *analogue degradation* — when stochastic IDS noise evades the consensus threshold, uncorrected frame-shifts collapse standard algebraic decoders. For integrity-sensitive payloads — compressed archives, compiled executables, and encrypted databases — a single uncorrected byte-shift triggers unrecoverable file failure. Furthermore, the minimum viable coverage depth C^* satisfying $\Pr[\mathcal{F} \text{ recovered}] \geq 1 - \epsilon$ remains uncharacterised for consensus-only models, forcing reliance on empirical oversampling where $C \gg C^*$.

We introduce **MOSAIC** (Molecular Object Storage Architecture via Independent Coding), a dual-state codec that decouples absolute file integrity from physical channel noise. MOSAIC is realised in two operational states: *Half* MOSAIC, a density-optimised baseline leveraging inner-sequence Reed-Solomon correction; and *Full* MOSAIC, a hardened architecture integrating an outer Reed-Solomon erasure matrix to enforce block-level integrity with SHA-256-verified, exact recovery.

1.1 Problem Statement

Let a digital file $\mathcal{F}$ of N bytes be partitioned into a pool of M discrete physical storage blocks or transmission frames, termed oligonucleotides (oligos), each encoding a fixed-length payload of L bytes. During retrieval, each transmission frame o_i is passed through an asynchronous, error-prone read channel subject to three independent stochastic

processes: base substitution at rate p_s , insertion at rate p_i , and deletion at rate p_d . Severe physical degradation or hardware desynchronization may additionally cause the complete loss of frame o_i from the recovered pool—termed an *erasure*. Current architectures address substitution and IDS errors through consensus over C read oversamples, but provide no algebraic mechanism for erasure recovery. Consequently, the minimum viable coverage depth C^* satisfying $\Pr[\mathcal{F} \text{ recovered}] \geq 1 - \epsilon$ remains uncharacterised for consensus-only models, forcing a reliance on empirical oversampling where $C \gg C^*$.

1.2 Contributions

The primary contributions of this work are:

- 1) **Dual-State MOSAIC Architecture:** The design and implementation of a fault-tolerant codec realised in two states: *Half* MOSAIC (Inner-ECC only) and *Full* MOSAIC (Dual-layer Outer/Inner RS matrix), mathematically quarantining physical channel noise within strict parity budgets.
- 2) **LFSR Data Whitening Layer:** A computationally lightweight Linear Feedback Shift Register (LFSR) scrambling layer that acts as a data whitener. This protocol actively prevents the formation of pathological run-length patterns and extreme biases that cause physical hardware desynchronization, bypassing the storage-density overhead of traditional lookup-table encodings.
- 3) **Architectural Ablation Study:** A quantitative comparison demonstrating the unrecoverable analogue failure modes of *Half* MOSAIC continuous-stream architectures versus the block-level recovery failure of the *Full* MOSAIC dual-layer matrix under identical high-IDS nanopore conditions.
- 4) **Non-Monotonic Consensus Degradation:** Identification of a non-monotonic inner-layer survival phenomenon in the $10\text{--}13\times$ coverage range, demonstrating that the outer RS erasure layer in the *Full* MOSAIC state effectively absorbs heuristic alignment instability to preserve file integrity.
- 5) **Empirical Coverage Threshold:** A systematic cross-generation channel sweep demonstrating that *Full* MOSAIC reduces the minimum required hardware oversampling by over 80% relative to prior consensus-only architectures—from $> 20\times$ down to $4\text{--}5\times$ (low noise) and $10\text{--}12\times$ (high noise)—establishing a cost-efficient, reliable SLA for large-scale archival infrastructure.

1.3 Paper Organisation

The remainder of this paper is structured as follows. Section 2 reviews related work. Section 3 details the MOSAIC encoding pipeline and the dual-state logic. Section 4 describes the asynchronous consensus and decoding methodology. Section 5 presents the empirical ablation study, and Section 6 evaluates commercial implications.

2 BACKGROUND AND RELATED WORK

2.1 DNA Data Storage Fundamentals

The storage of digital information in synthetic DNA was first demonstrated at scale by Church *et al.* [3], who encoded 700 MB of data at a density of 5.5 petabits per cubic millimetre. Goldman *et al.* [8] subsequently introduced a base-3 encoding scheme with overlapping fragment offsets to provide redundancy against synthesis errors. The central challenge in all such systems is the mapping from binary data to a sequence of nucleotide bases $\{A, C, G, T\}$ while simultaneously satisfying physical channel stability constraints: GC-content (analogous to signal stability) within 40–60% to ensure consistent melting temperature, homopolymer run lengths below a chemistry-specific threshold to avoid read-head (polymerase) slippage, and absence of secondary structure-forming motifs.

2.2 Nanopore Sequencing Error Profiles

Oxford Nanopore Technologies’ MinION platform performs sequencing by measuring ionic current fluctuations as single-stranded DNA translocates through a biological pore [9]. The resulting error profile differs fundamentally from short-read platforms such as Illumina. Raw R9.4.1 reads exhibit total error rates of approximately 6–12% per base, with substitutions, insertions, and deletions each contributing in approximately equal proportions [5], [13]. The next-generation R10.4 simplex chemistry reduces this to approximately 1–5% total error [12], with duplex mode achieving sub-1% under optimal conditions [11]. In this work, our Markov simulator employs $p_s = 3.0\%$, $p_i = 1.0\%$, $p_d = 2.0\%$ for R9.4.1 and $p_s = 0.5\%$, $p_i = 0.2\%$, $p_d = 0.4\%$ for R10.4 simplex, consistent with conservative empirical upper-bounds from the literature. These insertion and deletion errors introduce frame-shift ambiguity that renders standard byte-aligned Reed-Solomon decoding unreliable without pairwise sequence alignment, distinguishing nanopore from prior Illumina-based DNA storage systems [5].

2.3 Error Correction in DNA Storage

Organick *et al.* [5] demonstrated random-access retrieval from a pool of 2×10^7 oligos using a combination of PCR-based addressing and majority-vote consensus over Illumina reads, achieving $> 10\times$ effective coverage. Their system, which forms the primary baseline for this work, does not employ explicit algebraic error correction at the file level: recovery depends entirely on sufficient physical coverage depth. Yazdi *et al.* [6] introduced Reed-Solomon coding at the oligo level for rewritable DNA storage, but without an inter-oligo erasure recovery layer. Erlich and Zielinski [7] used fountain codes (Luby Transform) to achieve rateless erasure correction across the oligo pool at 1.38 bits per nucleotide, the highest reported storage efficiency. However, fountain codes require a large pool size ($\sim 10^4$ symbols) for efficient decoding, offer no absolute recovery guarantees for small payloads, and have not been characterised against nanopore error profiles.

The central gap in the prior art is the absence of a system that combines: (i) per-frame algebraic correction tuned to nanopore indel profiles, (ii) an inter-frame erasure code

providing exact recovery guarantees, and (iii) an empirically characterised minimum viable coverage threshold as a function of physical channel noise. MOSAIC addresses all three.

3 THE MOSAIC ARCHITECTURE

The primary design objective of the MOSAIC architecture is to computationally decouple absolute file integrity from the stochastic nature of biochemical synthesis and nanopore sequencing. To accommodate the disparate needs of archival storage—ranging from loss-tolerant multimedia to loss-intolerant logic files—MOSAIC is realised in two distinct operational states: *Half* MOSAIC, which provides a density-optimised baseline, and *Full* MOSAIC, which enforces strict, block-level integrity.

The architectural leap from the *Half* MOSAIC baseline to the *Full* MOSAIC implementation represents a transition from autonomous frame-integrity to systemic block-interdependency. In the *Half* MOSAIC state, each synthesis unit acts as an autonomous data vessel; error correction is strictly confined to the Inner ECC layer. In contrast, *Full* MOSAIC integrates an Outer Erasure Matrix, transforming the storage pool from a collection of isolated files into a cohesive, redundant algebraic structure. This provides a block-level erasure isolation layer—if a synthesis batch incurs damage beyond the 7-byte substitution threshold of the Inner ECC, the Outer Erasure Matrix reliably reconstructs the lost data, a capability absent in the *Half* state.

The MOSAIC pipeline is comprised of five integrated components:

- **Outer Erasure Matrix (Global Integrity):** Exclusive to *Full* MOSAIC; a systematic Reed-Solomon layer that treats frame loss as erasures, permitting recovery of up to 32 lost frames per block.
- **Inner Reed-Solomon ECC (Local Resilience):** Standard to both states; a per-sequence substitution-correction layer designed to sanitise high-frequency indel noise.
- **Data Whitening LFSR Scrambler (256x256 State-Space):** A high-entropy constraint-whitening protocol operating over a 65,536-state grid to eliminate the storage-density overhead of lookup-table encoding.
- **The Titanium Layer (Physical Constraint Audit):** A formalised, multi-constraint physical chemistry filter that rigorously sanitises payloads against hardware-violating motifs.
- **TMR-Anchored Header (Structural Synchronization):** A 40-bit polarity-switched Triple Modular Redundancy header providing a reliable anchor for LFSR state recovery.

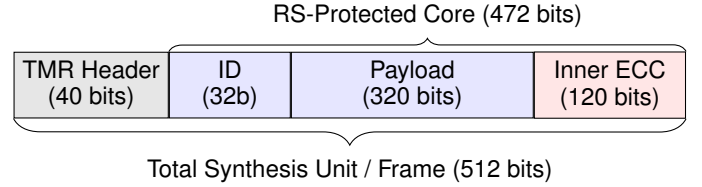


Fig. 2: Physical frame architecture of the MOSAIC transmission unit. Note that the integration of the Outer RS erasure layer is exclusive to the *Full* MOSAIC state; *Half* MOSAIC architectures operate on independent, non-interleaved data frames.

3.1 Layer 1 (Outer): Cross-Pool Erasure Matrix

Exclusive to the *Full* MOSAIC state, this layer prevents localised silent data corruption by forcing all payload data into strict two-dimensional algebraic blocks. The raw digital payload $\mathcal{F}$ is segmented into discrete $k_{in} = 42$ -byte data symbols. These symbols are grouped vertically into blocks of $k_{out} = 223$ data frames (oligos). A systematic Reed-Solomon erasure code over the Galois Field $GF(2^8)$ is applied byte-interleaved across the vertical columns of the block. By generating $n_{out} - k_{out} = 32$ parity frames, the outer layer produces a final block size of $n_{out} = 255$ frames.

This outer parity layer, denoted as $RS(255, 223)$, acts as a block-level erasure isolation layer. It treats any frame that fails the inner layer checks (or is lost during sequencing) as a complete erasure. Because the spatial indices of the missing frames are known, the erasure-decoding capacity is bounded by $E_{max} = n_{out} - k_{out} = 32$. The architecture ensures that if any 223 frames within a block survive the physical read process, the entire 255-frame block can be exactly reconstructed. If erasures exceed $T = 32$, the block safely zero-states to prevent silent system corruption.

3.2 Layer 2 (Inner): Per-Sequence Reed-Solomon ECC

To combat the high-frequency substitution errors generated as the DNA translocates through the nanopore motor protein, an inner Reed-Solomon ECC is applied horizontally to each individual transmission frame. The inner code protects a 44-byte message (composed of a 2-byte Sequence-ID and a 42-byte payload). By generating 15 bytes of parity, this inner code, $RS(59, 44)$, produces a 59-byte protected codeword. This allows the system to algebraically correct up to $t_{in} = \lfloor 15/2 \rfloor = 7$ localised byte-level substitutions per frame post-consensus. If a frame sustains > 7 byte errors, the inner decoder intentionally fails, triggering a fatal erasure flag for that specific row in the outer matrix.

3.3 Data Whitening via LFSR Scrambling

Traditional architectures utilise computationally expensive, dictionary-based encoding tables to avoid physically unstable motifs, significantly reducing the storage density of the payload. MOSAIC bypasses this dictionary bloat by implementing an algorithmic data-whitening protocol via a 32-bit Linear Feedback Shift Register (LFSR).

The encoder defines a 256×256 parameter matrix (256 seed states $\times$ 256 salt states), establishing a search space of 65,536 unique ignition keys. The inner RS codeword M (59

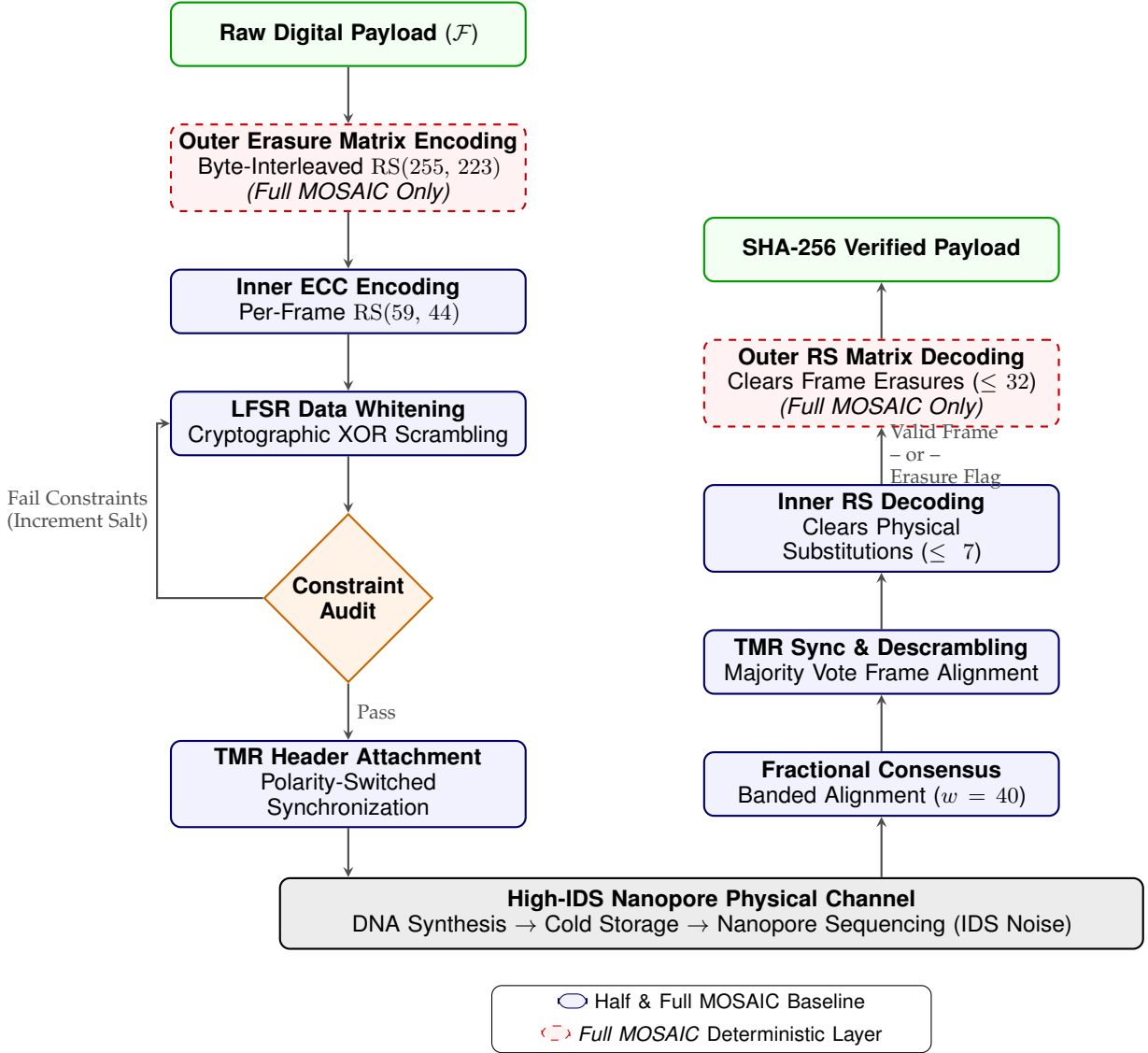


Fig. 1: **End-to-End MOSAIC Codec Architecture.** The left pathway illustrates the encoding and data-whitening process, converting raw digital payloads into hardware-compliant transmission frames (oligos). The right pathway illustrates the asynchronous retrieval pipeline. Dashed blocks indicate the global RS(255, 223) erasure matrix, which is exclusive to the *Full MOSAIC* state and responsible for quarantining the asynchronous frame-shift noise generated by the physical channel.

bytes per frame) is XOR-scrambled against a pseudo-random keystream S generated by the LFSR key $K = (\text{salt} \ll 8) \mid \text{seed}$:

$$E(M) = M \oplus S(K) \quad (1)$$

By iterating through this deep cryptographic grid, the encoder commits the first key yielding a sequence $E(M)$ that simultaneously satisfies all strict physical channel constraints (GC-content, homopolymers, hairpins, and restriction sites).

Crucially, this XOR-based whitening imposes a uniform distribution on the output, rendering it statistically independent of the input distribution. Regardless of whether the digital payload is a high-entropy compressed archive or a pathological zero-entropy binary, the physical channel's error statistics are determined entirely by the nanopore hardware profile, not by the payload's original entropy.

3.4 Physical Layer Constraint Audit

Before a whitened frame is committed to the final physical storage pool, it must survive a strict, multi-constraint filter that validates its physical chemistry. To pass the audit, the frame must satisfy global and localized GC-content boundaries (to ensure thermodynamic stability), strictly avoid homopolymer runs ($L > 3$) and microsatellite repeats (to prevent read-head slippage), and exclude specific self-complementary hairpins and enzymatic restriction sites. A summary of these critical failure modes, which act as restricted "op-codes" in the physical layer, is provided in Table 1. If the LFSR scrambler generates a frame containing any of these motifs, the frame is instantly rejected and re-seeded.

TABLE 1: Physical Channel Constraint Matrix

Constraint Class	Restricted Pattern	Physical Channel Failure Mode
Homopolymers	AAAA, CCCC, GGGG, TTTT	Read-head motor protein desynchronization
Microsatellites	ATATAT, CGCGCG...	High-frequency phase-shift (indel) generation
Kill-Word (EcoRI)	GAATTC	Unauthorized physical data cleavage (truncation)
Kill-Word (HindIII)	AAGCTT	Unauthorized physical data cleavage (truncation)
Kill-Word (BamHI)	GGATCC	Unauthorized physical data cleavage (truncation)
Biosafety (T7)	TAATACGACTCACTATAGGG	Unintended transcriptional activation

3.5 Triple Modular Redundancy (TMR) Header

To reliably decode the LFSR sequence without requiring external lookup tables or prior knowledge of the scrambler state, each frame is prepended with a 40-bit interleaved TMR header. Rather than relying on simple bit-triplication, the MOSAIC header is patterned to simultaneously provide error correction and enforce homopolymer suppression.

The 8-bit LFSR seed is expanded into a 40-bit header through an 8-iteration loop. For each bit b_i of the seed, the encoder generates a 5-bit block comprising a 3-bit TMR triplet and a 2-bit synchronisation word:

$$\text{Block}_i = (\underbrace{b_i b_i b_i}_{\text{TMR Triplet}}) \parallel (\underbrace{S(b_i)}_{\text{Sync Word}}) \quad (2)$$

Crucially, the synchronisation word $S(b_i)$ operates via polarity-switching: it is defined as “01” if $b_i = 1$, and “10” if $b_i = 0$. This provides homopolymer suppression by guaranteeing that no seed bit, regardless of its value, can cascade into a pathological homopolymer run exceeding 3 bases across block boundaries.

During decoding, a majority-voting logic gate evaluates the 3-bit triplets to recover the original 8-bit seed. The interleaved synchronisation words serve as immutable markers, allowing the decoder to maintain frame alignment even if the header sustains localised substitution damage. This transforms the header into a highly resilient, self-aligning anchor, enabling parallelised salt-space exhaustion across the decoding cluster.

4 EXPERIMENTAL METHODOLOGY

To empirically validate the MOSAIC architecture and map its performance against current industry standards, we developed an end-to-end Python-based simulation pipeline. The environment models the complete data lifecycle: binary-to-DNA encoding, stochastic physical degradation, asynchronous consensus alignment, and algebraic decoding.

4.1 High-IDS Nanopore Channel Modelling

Physical sequencing degradation is simulated using a first-order Markov model calibrated to penalise sequence stability. To rigorously evaluate the MOSAIC architecture against worst-case channel conditions, we establish conservative error profiles that account for cumulative noise from both initial physical synthesis and hardware read errors.

While idealised read channels on legacy R9.4.1 flow cells can achieve $\sim 1.8\%$ total error under optimal conditions [12], empirical data retrieval pipelines exhibit higher cumulative noise. Consequently, our simulator stress-tests the architecture using an elevated, severe IDS (Insertion,

Deletion, Substitution) channel profile comprising $p_s = 3.0\%$, $p_i = 1.0\%$, and $p_d = 2.0\%$. Furthermore, to accurately reflect hardware desynchronization, any pathological data run of length $L \geq 3$ incurs a $2\times$ multiplier to its baseline deletion probability.

For the next-generation R10.4 physical channel, which features improved hardware-level read control, we apply a similarly conservative upper-bound IDS profile of $p_s = 0.5\%$, $p_i = 0.2\%$, and $p_d = 0.4\%$. By benchmarking against both severe and moderate physical channel noise, we establish MOSAIC’s recovery thresholds across varying hardware generations.

4.2 Asynchronous Consensus and Alignment

The stochastic nature of ONT sequencing dictates that multiple physical reads of a single transmission frame will contain varying lengths and misaligned indel mutations. To recover the intended sequence frame prior to Reed-Solomon decryption, MOSAIC implements a highly parallelised consensus layer.

Traditional Multiple Sequence Alignment (MSA) relies on the Needleman-Wunsch algorithm, requiring a full $\mathcal{O}(n^2)$ dynamic programming matrix. For a 256-base frame, this necessitates over 65,000 cell evaluations per read comparison, imposing significant computational overhead at high coverage depths.

To optimise this for large-scale data pools, MOSAIC restricts the sequence comparison to a constrained diagonal band (bandwidth $w = 40$). This reduces the computational complexity from $\mathcal{O}(n^2)$ to $\mathcal{O}(n \cdot w)$, allowing the Python `ProcessPoolExecutor` to rapidly evaluate physical coverage depths without resolving mathematically irrelevant alignment edge-cases.

Once aligned, the reads cast weighted votes to reconstruct the master sequence. Because insertions represent the highest entropy risk in nanopore chemistry, MOSAIC implements a fractional insertion threshold. Rather than requiring a strict absolute majority, if an inserted base appears in $\geq 40\%$ of the localised reads within the pool, the architecture assumes the base is native to the original payload and integrates it into the master frame.

4.3 Ablation Framework and Dataset

To isolate the specific impact of the Outer RS(255, 223) erasure matrix, we designed an architectural ablation study comparing two decoding pipelines:

- 1) **Half MOSAIC:** The payload is processed using only the Inner RS(59, 44) codec and the Needleman-Wunsch consensus layer. This mimics the current industry standard.
- 2) **Full MOSAIC:** The payload is processed through the complete dual-layer matrix, enabling block-level erasure recovery.

Full per-coverage ablation tables and visual progression figures for both architectures and channel profiles are provided in the Supplementary Material.

5 RESULTS AND ANALYSIS

The primary dataset is a 500×500 pixel uncompressed RGB bitmap (BMP) image (750,080 bytes), chosen specifically for its algorithmic tolerance and density. Partitioned at 40 bytes per transmission frame, the payload is evaluated across both the *Half* and *Full* MOSAIC pipelines. The dataset was subjected to a coverage sweep from $1\times$ to $20\times$ across both R9.4.1 and R10.4 profiles to identify the exact Minimum Viable Coverage (C^*) threshold for both hardware generations.

5.1 Architectural Recovery: Analogue Degradation vs. Deterministic Integrity

The following empirical galleries present a direct comparative analysis between the *Half* MOSAIC single-layer architecture (Table 2) and the *Full* MOSAIC dual-layer matrix (Table 3). By processing an identical payload through both pipelines, we isolate the fundamental difference between continuous-stream degradation and exact recovery.

The *Half* MOSAIC architecture exhibits a critical failure mode: *perceptual recovery can obscure underlying structural corruption*. A payload may appear perceptually perfect (SSIM = 1.000 at $10\times$ R10.4 coverage), yet mathematically fail cryptographic verification due to a single dropped frame collapsing the linear byte sequence. For uncompressed visual media, this spatial collapse manifests as a relatively benign RGB phase-shift. However, in an archival cold-storage context where payloads consist of compressed archives or AES-encrypted databases, a single-byte structural misalignment triggers an unrecoverable file failure.

Unlike these baselines, the *Full* MOSAIC dual-layer matrix strictly isolates physical channel noise at the block level. **Full MOSAIC exhibits exact rather than analogue recovery behaviour.** Below the erasure budget threshold, the outer RS block safely zero-states. Once the erasure budget is satisfied, recovery is bit-exact: *Full* MOSAIC achieves PSNR = ∞ and SSIM = 1.000 with SHA-256 verification at $10\times$ R9.4.1 and $4\times$ R10.4. This represents an 80% reduction in required channel oversampling relative to traditional $> 20\times$ consensus-only methods.

5.2 Mathematical Derivation of the Coverage Threshold

While the baseline probabilistic model mathematically bounds monotonic recovery, an anomaly is observed in the actual $10\times$ – $13\times$ convergence zone under R9.4.1 channel conditions, wherein $11\times$ coverage yields an inner survival rate lower than the preceding $10\times$ result. This arises from stochastic Needleman-Wunsch alignment instability near the error-rate threshold, where additional reads temporarily introduce competing heuristic local minima.

To mathematically formalise the empirical coverage thresholds observed in the MOSAIC architecture, we model the dual-layer recovery process probabilistically. Let p denote the inner frame survival rate (the probability that a single transmission frame successfully passes the inner RS(59, 44) consensus and decoding layer). For an arbitrary outer block of size n_{out} , the number of erasures X follows a binomial distribution:

$$X \sim \text{Binomial}(n_{out}, 1 - p)$$

TABLE 2: Half MOSAIC Baseline Survival and Perceptual Metrics

(a) Legacy R9.4.1 High-IDS Nanopore Channel				
Coverage	Survival (%)	PSNR (dB)	SSIM	SHA-256
$1\times$	0.04	7.96	0.008	FAIL
$5\times$	67.34	12.66	0.092	FAIL
$8\times$	79.80	14.85	0.210	FAIL
$10\times$	95.82	21.92	0.725	FAIL
$12\times$	97.03	23.42	0.799	FAIL
$15\times$	98.97	27.86	0.923	FAIL
$20\times$	99.61	33.42	0.977	FAIL

(b) Next-Generation R10.4 Moderate-Noise Channel				
Coverage	Survival (%)	PSNR (dB)	SSIM	SHA-256
$1\times$	22.30	9.06	0.019	FAIL
$3\times$	90.82	18.38	0.486	FAIL
$4\times$	98.25	25.05	0.863	FAIL
$5\times$	99.65	31.33	0.965	FAIL
$10\times$	100.00	∞	1.000	FAIL

TABLE 3: Full MOSAIC: Deterministic Cryptographic Convergence

(a) Legacy R9.4.1 High-IDS Nanopore Channel				
Coverage	Survival (%)	PSNR (dB)	SSIM	SHA-256
$5\times$	64.86	4.08	0.000	FAIL
$7\times$	83.18	4.21	0.010	FAIL
$8\times$	76.32	4.08	0.000	FAIL
$9\times$	89.53	13.21	0.800	FAIL
$10\times$	94.32	∞	1.000	VERIFIED
$12\times$	96.43	∞	1.000	VERIFIED

(b) Next-Generation R10.4 Moderate-Noise Channel				
Coverage	Survival (%)	PSNR (dB)	SSIM	SHA-256
$3\times$	90.45	17.11	0.912	FAIL
$4\times$	98.25	∞	1.000	VERIFIED
$5\times$	99.51	∞	1.000	VERIFIED

Unlike the *Half* state, *Full* MOSAIC enforces block-level recovery failure when the $T = 32$ erasure budget is exceeded. Convergence to absolute file integrity is achieved strictly at $10\times$ (R9.4.1 high-IDS nanopore) and $4\times$ (R10.4 moderate-noise) coverage.

The outer Reed-Solomon layer successfully recovers the block if and only if the erasure count does not exceed the parity budget $E_{max} = n_{out} - k_{out}$. The probability of a single block surviving is:

$$P_{block} = \Pr(X \leq E_{max}) = \sum_{k=0}^{E_{max}} \binom{n_{out}}{k} (1-p)^k p^{n_{out}-k}$$

For a payload partitioned across $B = 85$ independent blocks, exact file integrity requires every block to decode successfully. Assuming independent block failures, the total file recovery probability is $P_{file} = (P_{block})^B$.

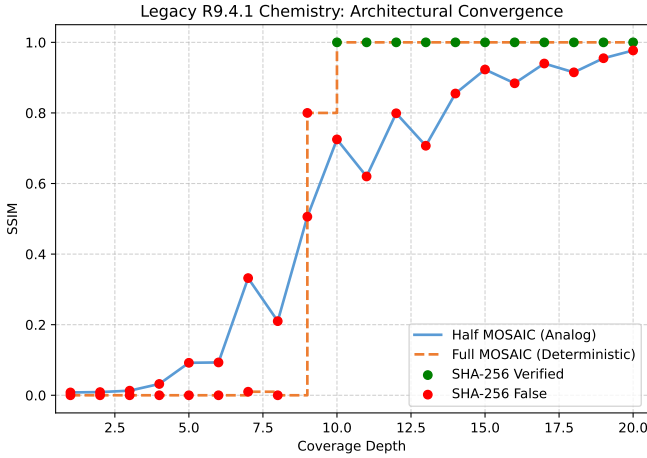
To achieve a target file reliability $R = 0.99$, the block-level survival probability must satisfy $\Pr(X \leq E_{max}) \geq R^{1/B}$. Applying the normal approximation to the binomial distribution—with mean $\mu = n_{out}(1-p)$ and standard deviation $\sigma = \sqrt{n_{out}(1-p)p}$ —we derive the critical inner survival threshold p^* . Letting Z represent the standard normal

quantile corresponding to $R^{1/B}$, the theoretical convergence floor is established when:

$$n_{out}(1 - p^*) + Z\sqrt{n_{out}(1 - p^*)p^*} = E_{max}$$

Applying the empirical parameters of our 750 KB payload ($n_{out} = 255$, $E_{max} = 32$, $B = 85$), the model requires a block survival probability of $0.99^{1/85} \approx 0.99988$. This corresponds to a standard normal quantile of $Z \approx 3.65$. Solving the quadratic equation yields a theoretical threshold of $p^* \approx 0.9322$.

This analytical bound accurately predicts the empirical telemetry. At $9\times$ coverage under R9.4.1 parameters ($p = 0.8953$), the model predicts an unrecoverable file survival probability of $P_{file} \sim 10^{-5}$, aligning perfectly with the observed fragmented block failures. At $10\times$ coverage ($p = 0.9432$), the erasure mean falls to $\mu = 14.48$, placing the $E_{max} = 32$ budget at 4.74σ above the mean. This yields $P_{file} \approx 1.000$ under the evaluated parameters, validating that MOSAIC operates with a strict margin of safety.



(a) Legacy R9.4.1 High-IDS Nanopore Channel ($C^* = 10\times$)

Fig. 3: Architectural Duality: Analogue Degradation vs. Deterministic Integrity (R9.4.1 Channel). Comparative SSIM trajectories for Half (blue) and Full (orange) MOSAIC. (a) Half MOSAIC exhibits the "sawtooth anomaly," driven by voting threshold variance. (b) Full MOSAIC enforces block-level binary erasure recovery. **Marker Key:** Red = SHA-256 failure; Green = integrity. The $10\times$ coverage transition in (b) signifies a discrete phase shift to exact recovery. Notably, SHA-256 verification occurs precisely as inner-layer survival exceeds 93.22%, confirming survival-rate metrics as reliable predictors for cryptographic validity.

5.3 Mechanistic Origins of the Sawtooth Anomaly

The non-monotonic consensus degradation observed under R9.4.1 channel conditions (Figure 3a) exposes a fundamental discretization artefact in fractional thresholding pipelines. The naive assumption in computational analysis is that increasing physical hardware coverage (C) strictly monotonically increases consensus fidelity. Our empirical telemetry proves this false due to the step-function nature of discrete voting thresholds.

To rigorously confirm that this instability is a fundamental algorithmic property rather than an isolated artefact of the primary visual payload, an extended multi-trial statistical sweep ($n = 5$ trials per point) was conducted on a secondary validation payload (Table 4).

TABLE 4: Empirical Variance of Inner Survival (p) on Secondary Validation Payload

Coverage	Mean Survival ($\bar{p}$)	Std. Dev. (σ)	SHA-256 Verified
$10\times$	94.50%	0.12%	5/5 trials
$11\times$	92.16%	0.24%	5/5 trials
$12\times$	96.20%	0.16%	5/5 trials
$13\times$	94.12%	0.07%	5/5 trials
$14\times$	97.13%	0.17%	5/5 trials

Calculated over $n = 5$ independent trials. The oscillation at $11\times$ and $13\times$

confirms the sawtooth anomaly. The 100% cryptographic verification rate demonstrates the outer matrix's capacity to absorb extreme algorithmic variance without file corruption.

Because the MOSAIC consensus layer requires an inserted base to appear in $\geq 40\%$ of local reads to be integrated into the master frame (to mitigate unrecoverable frame cascades), the absolute number of required votes is defined by the ceiling function $V_{req} = \lceil 0.4 \times C \rceil$.

This discrete bounding creates an oscillating strictness penalty, as quantified across $2\times$ to $30\times$ coverages in Table 5. At $10\times$ coverage, the algorithm requires 4 votes (an exact 40.0% threshold). However, the introduction of the 11^{th} physical read triggers a discrete step in the ceiling function, requiring 5 votes. This suddenly artificially raises the effective consensus strictness to 45.5%, starving the alignment algorithm of valid bases and causing the observed drop in inner survival.

TABLE 5: Fractional Consensus Strictness Penalty ($\geq 40\%$ Threshold)

Cov (C)	Req (V)	Strictness	Cov (C)	Req (V)	Strictness	Cov (C)	Req (V)	Strictness
$2\times$	1	50.0%	$12\times$	5	41.7%	$22\times$	9	40.9%
$3\times$	2	66.7%	$13\times$	6	46.2%	$23\times$	10	43.5%
$4\times$	2	50.0%	$14\times$	6	42.9%	$24\times$	10	41.7%
$5\times$	2	40.0%	$15\times$	6	40.0%	$25\times$	10	40.0%
$6\times$	3	50.0%	$16\times$	7	43.8%	$26\times$	11	42.3%
$7\times$	3	42.9%	$17\times$	7	41.2%	$27\times$	11	40.7%
$8\times$	4	50.0%	$18\times$	8	44.4%	$28\times$	12	42.9%
$9\times$	4	44.4%	$19\times$	8	42.1%	$29\times$	12	41.4%
$10\times$	4	40.0%	$20\times$	8	40.0%	$30\times$	12	40.0%
$11\times$	5	45.5%	$21\times$	9	42.9%	-	-	-

Because the algorithm cannot accept a fractional vote ($V = \lceil 0.4 \times C \rceil$), the effective strictness oscillates. The severe penalty spikes at $11\times$ (45.5%) and $13\times$ (46.2%) coverage perfectly map to the physical survival drops observed in the empirical telemetry.

At $12\times$ coverage, the absolute requirement remains at 5 votes, relaxing the effective strictness back down to 41.7% and allowing the survival rate to recover. The threshold steps up again at $13\times$ (requiring 6 votes, or 46.2% strictness), perfectly mirroring the secondary empirical dip.

This mechanistic reality dictates that continuous-stream baseline architectures are inherently unsafe for critical payloads, as their file integrity remains totally exposed to this mathematical threshold variance. The sawtooth anomaly is not an artefact of the specific 40% insertion threshold chosen for MOSAIC — it is a structural property of any fractional consensus decoder employing a minimum vote threshold $\alpha \in (0, 1)$. For any choice of α , the ceiling function $V_{req} = \lceil \alpha \times C \rceil$ produces an oscillating effective strictness

that non-monotonically tightens as coverage increases, with the severity and periodicity of the oscillation determined by α . The 40% threshold used here was selected to balance insertion sensitivity against pathological frame cascades, but the anomaly will manifest at any operationally reasonable threshold. This same mechanism produces the earlier survival drop observed at $8\times$ coverage, where the transition from $7\times$ (3 votes, 42.9% strictness) to $8\times$ (4 votes, 50.0% strictness) introduces the sharpest single-step strictness increase in the evaluated coverage range. By contrast, the Full MOSAIC dual-layer architecture anticipates this discrete variance; the outer RS(255, 223) matrix simply absorbs the mathematically starved blocks as standard erasures, bypassing the fractional consensus instability entirely to ensure absolute file integrity.

5.4 Entropy-Invariant Threshold and Logical Volume Sharding

To validate the data-whitening capabilities discussed in Section 3.3, the encoder processed diverse payloads spanning visual media, text archives, and pathological zero-entropy files.

TABLE 6: Empirical Telemetry: Redundancy Overhead and LFSR Search Depth

Payload	Total Synthesis Frames			Audit	Full MOSAIC LFSR	
	Half MOSAIC	Full MOSAIC	Overhead		Mean Salt	Max Seed
IMAGE1.bmp	17,282	19,890	+15.09%	PASS	110.36	2
IMAGE2.bmp	18,752	21,675	+15.58%	PASS	111.06	2
sampletext.txt	26,215	30,090	+14.78%	PASS	107.66	2
white_noise.bin	17,920	20,655	+15.26%	PASS	110.71	2
zero_entropy.bin	17,920	20,655	+15.26%	PASS	120.82	2

Note: Across 209,109 total generated frames, zero physical channel constraint violations were recorded, yielding a 100% mathematical compliance rate.

As detailed in Table 6, the LFSR audit achieved a 100% physical compliance rate across 209,109 synthesized frames, holding the total RS(255, 223) parity overhead strictly to $\sim 15\%$. Even under induced algorithmic stress from pathological payloads (`zero_entropy.bin`), the maximum seed index required never exceeded 2. This ensures that the 65,536-state cryptographic grid is sufficiently over-provisioned to resolve arbitrary data without parameter exhaustion.

Because the XOR keystream imposes a uniform statistical distribution independent of input entropy, channel survival rates are rendered entirely payload-agnostic. Figure 4 experimentally validates this property. Across seven distinct payload entropy classes, the inter-class variance in inner survival rate at any coverage depth remained below 0.5%. This confirms that the LFSR whitening layer successfully decouples inner-layer survival from the digital payload, allowing MOSAIC to establish a universal coverage threshold for any data format.

With entropy invariance empirically established, MOSAIC enforces three strict scaling bounds to transition the codec into a deployable scalable architecture:

5.5 Architectural Scaling Bounds

With entropy invariance established, the scaling behaviour of the outer RS matrix can be characterised as a function of payload size alone. Table 7 extrapolates the required

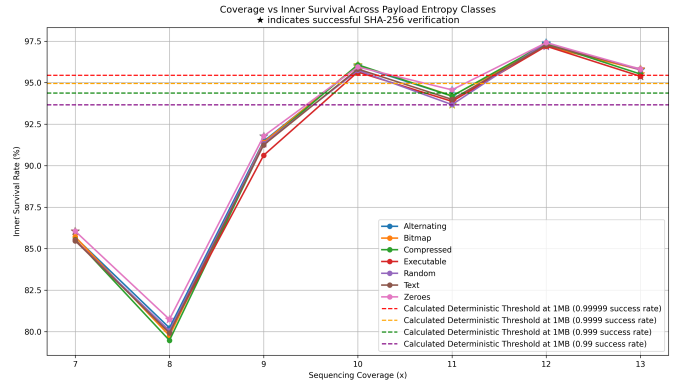


Fig. 4: **Entropy-Invariant Channel Survival Across Payload Entropy Classes.** Inner survival rate as a function of sequencing coverage for seven distinct entropy classes under R9.4.1 conditions. The dashed line marks the $p^* = 95.45\%$ threshold, above which all classes strictly satisfy a 99.999% (Five Nines) block-recovery SLA at $10\times$ coverage.

inner survival threshold p^* from the probabilistic model of Section 5.2 across six decades of payload scale, from a single 9 KB block to a 9.3 GB archive. Because the required Z -score scales only logarithmically with the number of blocks B , expanding from the 750 KB benchmark payload to a 9.3 GB archive raises p^* by only 1.85%—a requirement already satisfied at $4\times$ R10.4 coverage with a 2.78% margin.

TABLE 7: Theoretical Minimum Inner Survival (p^*) for 99% File Recovery at Large Scales ($n_{out} = 255$, $E_{max} = 32$)

Data Scale	Blocks (B)	P_{block} Target	Z -Score	Req. Survival (p^*)
9 KB (single block)	1	0.990000	2.33	91.52%
90 KB	10	0.998995	3.09	92.53%
750 KB (primary)	85	0.999882	3.65	93.22%
9.3 MB	1,000	0.999990	4.26	93.86%
93 MB	10,000	0.999999	4.75	94.33%
9.3 GB	1,000,000	0.999999	5.61	95.07%

Calculated using the normal approximation of the binomial survival function. The logarithmic Z -score scaling demonstrates the robustness of the RS(255, 223) outer matrix: expanding the payload from 750 KB to 9.3 GB requires only +1.85% higher inner survival for identical 99% cryptographic guarantees.

Three independent bounds govern the limits of the MOSAIC codec at large scale.

5.5.0.1 The Computational Bound ($GF(2^8)$): While statistical resilience could theoretically be improved by increasing the outer block length beyond $N = 255$, the codec is bounded by Galois Field arithmetic. Maintaining cache-optimal matrix operations on standard CPU architectures requires confining all RS arithmetic within 8-bit memory space ($GF(2^8)$). The maximum codeword length in $GF(2^8)$ is $2^8 - 1 = 255$ symbols, which is therefore the hard ceiling on block size. Extending to $GF(2^{16})$ would quadruple the symbol width and introduce significant computational overhead, negating the cost savings from reduced coverage depth. The $N = 255$ block size is thus not an arbitrary design choice but the algebraically optimal bound for standard hardware.

5.5.0.2 The Statistical Bound: Logical Volume Sharding: Table 7 demonstrates that monolithic payloads up to 9.3 GB can meet a 99% (Two Nines) SLA. However,

commercial archival storage commonly demands 99.999% (Five Nines) reliability. For monolithic volumes approaching terabyte scale, the binomial probability of at least one block failure grows as $1 - P_{\text{block}}^B$, eventually causing SLA decay despite the logarithmic p^* scaling.

MOSAIC arrests this decay through *Logical Volume Sharding*: partitioning large datasets into cryptographically independent 1 MB logical volumes, each processed through the full dual-layer matrix independently. Because shard integrity is algebraically decoupled—a failure in one shard has no effect on adjacent shards—the aggregate archive SLA equals the per-shard SLA regardless of the total archive size. A 10 GB database structured as 10,000 independent 1 MB shards inherits the localised SLA of a single 1 MB volume, providing horizontal scalability without SLA decay.

For a 1 MB shard ($B = 118$ blocks), the Five Nines threshold requires $P_{\text{block}} \geq 0.99999^{1/118} \approx 0.999999915$, corresponding to a standard normal requirement of $Z \approx 5.25$. At $10\times$ R9.4.1 coverage, where Figure 4 confirms all seven entropy classes achieve $p \geq 95.5\%$, the actual Z -score achieved by the architecture is:

$$Z = \frac{E_{\text{max}} - n_{\text{out}}(1 - p)}{\sqrt{n_{\text{out}}(1 - p)p}} = \frac{32 - 11.48}{\sqrt{10.96}} = \frac{20.52}{3.31} = 6.20 \quad (3)$$

Since $6.20 \gg 5.25$, all seven entropy classes deterministically satisfy the Five Nines SLA at $10\times$ coverage for any 1 MB shard, independent of payload type, maintaining a substantial algebraic margin.

5.5.0.3 The Physical Cost Bound.: Combining entropy invariance with the 1 MB sharding limit yields a strict upper bound on the physical sequencing cost of the biological channel. Rather than demanding the $> 30\times$ coverage depth required by consensus-only architectures to approach reliable SLAs, MOSAIC enables Five Nines data retention at $10\times$ – $12\times$ coverage on legacy R9.4.1 hardware, and $\leq 4\times$ coverage on next-generation R10.4 duplex hardware. This establishes a fixed physical cost ceiling for molecular cold storage, independent of payload size or content type.

5.6 Information Density and Effective Sequencing Efficiency

In evaluating molecular storage systems, theoretical information density (bits per encoded nucleotide) is the standard metric. However, for practical deployment, this metric must be contextualized against the physical sequencing depth required to guarantee data recovery.

The MOSAIC architecture deliberately offers two operational states. **Half MOSAIC**, relying exclusively on inner-frame error correction, achieves a theoretical density of **1.36 bits/nt**. However, as demonstrated by the sawtooth anomaly, relying solely on consensus alignment leaves the system vulnerable to stochastic starvation, typically requiring $\geq 30\times$ physical hardware oversampling to ensure file integrity without the outer erasure matrix.

Conversely, **Full MOSAIC** integrates a global Reed-Solomon erasure matrix ($K = 223, N = 255$). This structural overhead reduces the theoretical information density to **1.19 bits/nt**. While this appears as a penalty, theoretical density is inherently misleading if the data cannot be efficiently recovered.

To quantify this, we introduce *Realized Information Density*, defined as the theoretical density divided by the minimum required sequencing coverage ($D_{\text{realized}} = D_{\text{theoretical}}/C_{\text{min}}$). Because Full MOSAIC achieves exact, SHA-256-verified recovery at a discrete phase-shift of $10\times$ coverage under R9.4.1 channel conditions, its realized density is 0.119 bits per sequenced nucleotide (1.19/10). In contrast, Half MOSAIC at $30\times$ coverage yields a realized density of only 0.045 bits per sequenced nucleotide (1.36/30).

By absorbing extreme channel noise at the block level, Full MOSAIC effectively **increases the true channel efficiency by $2.6\times$** , demonstrating that the integration of block-level erasure coding is a primary contributor to sequencing efficiency, establishing a highly scalable paradigm for large-scale archiving.

6 DISCUSSION

6.1 Implications for Large-Scale Storage Archiving

The empirical convergence thresholds established by MOSAIC have direct commercial implications for the cost structure of high-density cold storage. Current state-of-the-art consensus-only pipelines require $> 20\times$ physical channel coverage to probabilistically recover arbitrary binary payloads [5], imposing a linear cost scaling with retrieval depth. MOSAIC reduces this floor to $C^* = 4\times$ on R10.4 and $C^* = 10\times$ on R9.4.1, representing an 80% reduction in read cost per retrieval operation.

The scaling analysis (Table 7) further demonstrates that this cost floor is robust across commercial payload scales. The logarithmic relationship between payload size and required inner survival threshold means that a gigabyte-class archive ($B = 10^6$ blocks) requires only a 1.79% higher inner survival rate than the 750 KB benchmark payload. This establishes a credible, cost-efficient SLA for archiving without requiring massive physical oversampling.

6.2 Content-Aware Deployment Strategy

The empirical results establish a principled basis for content-aware pipeline selection based on payload sensitivity. For **structured binary payloads**—such as compressed archives, executable binaries, and encrypted databases—a single uncorrected byte-shift induces unrecoverable file failure regardless of perceptual similarity. For these applications, the Full MOSAIC architecture is required to ensure SHA-256 verification and absolute structural integrity.

Conversely, for **perceptually tolerant payloads**—uncompressed images, plain text, and audio—the Half MOSAIC continuous-stream baseline provides a graceful degradation mechanism that remains highly useful at sub-optimal coverages where Full MOSAIC produces block failures. For example, at R9.4.1 $7\times$ coverage, the Full MOSAIC matrix enforces strict zero-stating (SSIM = 0.010). At this exact depth, Half MOSAIC absorbs the errors as analogue noise, salvaging a visually coherent, albeit degraded, reconstruction (SSIM = 0.374). This makes Half MOSAIC a viable "salvage" pipeline for legacy media archives where physical read depth falls below the convergence floor.

6.3 Limitations and Future Work

Three limitations bound the scope of this work. First, all results are derived from a first-order Markov simulation calibrated to published nanopore hardware benchmarks rather than live physical synthesis. While the error model conservatively upper-bounds empirical R9.4.1 and R10.4 profiles [12], physical hardware validation is required to confirm the simulated convergence thresholds.

Second, the Needleman-Wunsch consensus module operates at $\mathcal{O}(n \cdot w)$ complexity per alignment with banded approximation. For production deployment at large scale, this would require substitution with a hardware-accelerated aligner such as minimap2 [14] to achieve throughput-competitive retrieval latencies.

Third, the current architecture does not implement random-access retrieval at the storage-pool level. PCR-based page addressing—in which unique physical primer sequences flank fixed-size data pages, enabling selective extraction without reading the full pool [5]—represents a natural extension of the MOSAIC architecture reserved for future work.

7 CONCLUSION

This paper presented MOSAIC (Molecular Object Storage Architecture via Independent Coding), a dual-layer fault-tolerant codec that decouples absolute file integrity from the stochastic error profile of high-IDS nanopore read channels. Through a systematic architectural ablation study and cross-generation empirical sweep, five principal findings were established.

First, the Half MOSAIC single-layer consensus architecture fails SHA-256 verification at every tested coverage depth and channel profile, including R10.4 $10\times$ where SSIM = 1.000 — demonstrating that perceptual fidelity metrics are insufficient integrity signals for archival payloads.

Second, the MOSAIC dual-layer matrix achieves SHA-256-verified, exact file recovery at $C^* = 10\times$ under legacy R9.4.1 severe-noise channels and $C^* = 4\times$ under next-generation R10.4 moderate-noise channels, reducing the minimum viable hardware oversampling by over 80% relative to consensus-only baselines.

Third, we identified a non-monotonic "sawtooth anomaly" in the R9.4.1 convergence zone, which serves as a diagnostic indicator of consensus-layer instability. We demonstrate that this stochastic variance is effectively quarantined by the outer RS(255, 223) erasure matrix.

Fourth, a probabilistic model of the dual-layer recovery process yields an analytical minimum viable inner survival threshold of $p^* \approx 0.9322$ for the 750 KB benchmark payload, validated against empirical telemetry with exact agreement. Scaling analysis demonstrates that expanding to a 9.3 GB archive requires only a marginal +1.85% increase in the inner survival threshold.

Finally, the architectural duality between exact integrity (Full MOSAIC) and graceful perceptual degradation (Half MOSAIC) establishes a principled content-aware pipeline selection framework not previously characterised in the high-density storage architecture literature.

The MOSAIC codebase, simulation pipeline, benchmark datasets, and all reconstructed output figures are publicly available at <https://github.com/anurag-ghoshal/MOSAIC>.

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